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Harvesting Loss Detection Location and Improvement System and Method

Abstract

An agricultural machine for reducing harvesting loss during a harvesting operation includes: front ground engaging mechanisms coupled to a front axle, a chassis supported above the ground by the front ground engaging mechanisms, a cutting head located forward of the front ground engaging mechanisms and configured to harvest crop in a worksite, an image sensor, and a controller operatively coupled to the image sensor. The image sensor captures images of a field of view, which includes an area rearward of the front axle. The controller receives data corresponding to the images from the image sensor, determines the amount of grain shown in the images, and adjusts an operational characteristic of the agricultural machine based on the amount of grain determined to be shown in the images.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present description relates to agricultural machines and, in particular, to systems and methods for reducing harvesting loss of agricultural machines.

BACKGROUND OF THE DISCLOSURE

[0002] There are a variety of different types of agricultural machines. Some agricultural machines include combine harvesters, sugar cane harvesters, cotton harvesters, self-propelled forage harvesters, and windrowers. During harvesting operations, agricultural machines may have harvesting losses. Accurately detecting, categorizing, and reducing harvesting losses may be beneficial for various applications.

SUMMARY

[0003] In an illustrative implementation, an agricultural machine for reducing harvesting loss during a harvesting operation comprises: front ground engaging mechanisms coupled to a front axle; rear ground engaging mechanisms coupled to a rear axle; a chassis supported above a surface by the front ground engaging mechanisms and rear ground engaging mechanisms; a cutting head located forward of the front ground engaging mechanisms and configured to harvest crop in a worksite; at least one image sensor configured to capture one or more images of a field of view, the field of view being external to the agricultural machine and including an area rearward of a centerline of the front axle; and a controller configured to receive data corresponding to one or more images of the field of view from the at least one image sensor, determine the amount of grain shown in the one or more images of the field of view based on the received data, and adjust at least one operational characteristic of the agricultural machine based on the amount of grain determined to be shown in the one or more images of the field of view. In some implementations, the field of view includes an area underneath a portion of the chassis.

[0004] In some implementations, the agricultural machine further comprises: at least one light emitting device configured to emit light into the field of view. In some implementations, the at least one light emitting device includes at least one laser. In some implementations, the controller is configured to adjust at least one of the following operational characteristics of the agricultural machine based on the amount of grain determined to be shown in the one or more images of the field of view: speed of the agricultural machine, direction of travel of the agricultural machine, at least one operational characteristic of the cutting head, and at least one operational characteristic of a threshing assembly of the agricultural machine that is configured to process crop harvested by the cutting head.

[0005] In some implementations, the field of view includes an area forward of a center line of the rear axle. In some implementations, the agricultural machine further comprises: a threshing assembly positioned rearward of the cutting head; and a clean crop routing assembly configured to cooperate with the threshing assembly to separate grain from material other than grain of the harvested crop; and the at least one image sensor is located underneath at least one of the threshing assembly and the clean crop routing assembly. In some implementations, the at least one image sensor is directed laterally inward toward a lateral centerline of the agricultural machine.

[0006] In some implementations, the agricultural machine further comprises: a threshing assembly positioned rearward of the cutting head; a clean crop routing assembly configured to cooperate with the threshing assembly to separate grain from material other than grain of the harvested crop; and side walls between which the threshing assembly and the clean crop routing assembly are positioned; wherein the at least one image sensor is located on at least one of the side walls. In some implementations, the cutting head extends laterally from a first end to a second end; and the at least one image sensor is located on at least one of the first end and the second end of the cutting head.

[0007] In some implementations, the agricultural machine further comprises: a threshing assembly positioned rearward of the cutting head; a clean crop routing assembly configured to cooperate with the threshing assembly to separate grain from material other than grain of the harvested crop; a spreader configured to output the material other than grain from the agricultural machine; and an additional image sensor configured to capture one or more images of an additional field of view that includes an area configured to receive the material other than grain of the harvested crop output from the agricultural machine.

[0008] In some implementations, the controller is configured to: receive an indication of a pre-harvest grain loss; and generate a cutting head harvesting loss map, which indicates grain loss that has occurred during the harvesting operation and one or more corresponding locations in the worksite where the grain loss has occurred during the harvesting operation, based on the indication of pre-harvest grain loss and based on the amount of grain determined to be shown in the one or more images of the field of view.

[0009] In another illustrative implementation, an agricultural machine for reducing harvesting loss during a harvesting operation comprises: front ground engaging mechanisms configured to rotate during movement of the agricultural machine; rear ground engaging mechanisms configured to rotate during movement of the agricultural machine; a chassis supported above a surface by the pair of front ground engaging mechanisms and the pair of rear ground engaging mechanisms; a cutting head located forward of the pair of front ground engaging mechanisms and configured to harvest crop; a slope conveyor located rearward of the cutting head and forward of an inlet of a threshing assembly, the threshing assembly being configured to process harvested crop; at least one image sensor configured to capture one or more images of a field of view that is external to the agricultural machine and includes an area underneath the threshing assembly; and a controller configured to receive data corresponding to one or more images of the field of view from the at least one image sensor and determine the amount of grain shown in the one or more images of the field of view based on the received data corresponding to one or more images of the field of view.

[0010] In some implementations, the controller is configured to adjust at least one operational characteristic of the cutting head based on the amount of grain shown in the one or more images of the field of view. In some implementations, the controller is configured to adjust at least one operational characteristic of the threshing assembly based on the amount of grain shown in the one or more images of the field of view.

[0011] In some implementations, the field of view includes an area forward of the rear ground engaging mechanisms. In some implementations, the agricultural machine further comprises: a clean crop routing assembly configured to cooperate with the threshing assembly to separate grain from material other than grain of the harvested crop; and a first side wall and a second side wall between which the threshing assembly and the clean crop routing assembly are positioned; wherein the at least one image sensor is located on at least one of the first side wall and the second side wall.

[0012] In some implementations, the controller is configured to: receive an indication of a pre-harvest grain loss; and determine a cutting head grain loss value based on the indication of pre-harvest grain loss and based on the amount of grain determined to be shown in the one or more images of the field of view.

[0013] In another illustrative implementation, a method for reducing harvesting loss of an agricultural machine during a harvesting operation comprises: capturing one or more images of a field of view that is external to the agricultural machine and includes an area underneath a threshing assembly of the agricultural machine, the threshing assembly being configured to process harvested crop received from a slope conveyor located rearward of a cutting head that harvests crop and forward of an inlet of the threshing assembly; receiving, via a controller, data corresponding to one or more images of the field of view from the at least one image sensor; and determining, via

the controller, the amount of grain shown in the one or more images of the field of view based on the received data corresponding to one or more images of the field of view.
[0014] In some implementations, the method further comprises: adjusting an at least one operational characteristic of the agricultural machine based on the amount of grain determined, via the controller, to be shown in the one or more images of the field of view.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0015] The above-mentioned aspects of the present disclosure and the manner of obtaining them will become more apparent and the disclosure itself will be better understood by reference to the following description of the implementations of the disclosure, taken in conjunction with the accompanying drawings, wherein:
[0016] FIG. **1** is a side view of an example agricultural machine configured to harvest and process crop in a worksite;
[0017] FIG. **2***a* is a perspective view of an example cutting head for the agricultural machine;
[0018] FIG. **2***b* is a perspective view of another example cutting head for the agricultural machine;
[0019] FIG. **3** is a diagrammatic view of an example control system for the agricultural machine configured to measure and analyze data associated with harvested crop in the worksite and cause adjustments to operational characteristics of the agricultural machine based on the measured and analyze data;
[0020] FIG. **4** is a flow diagram showing an example method associated with reducing cutting head harvesting loss of the agricultural machine;
[0021] FIG. **5** is a flow diagram showing an example calibration method, for an exemplary light emission variable, the calibration method being associated with reducing cutting head harvesting loss of the agricultural machine;
[0022] FIG. **6** is a flow diagram showing an example calibration method, applicable to a plurality of light emission variables, the calibration method being associated with reducing cutting head harvesting loss of the agricultural machine; and
[0023] FIG. **7** is a flow diagram showing another example method associated with reducing cutting head harvesting loss of the agricultural machine.
[0024] Corresponding reference numerals are used to indicate corresponding parts throughout the several views.
DETAILED DESCRIPTION
[0025] The implementations of the present disclosure described below are not intended to be exhaustive or to limit the disclosure to the precise forms in the following detailed description. Rather, the implementations are chosen and described so that others skilled in the art may appreciate and understand the principles and practices of the present disclosure.
[0026] In FIG. **1**, an implementation of an agricultural machine **10** is shown. The agricultural machine **10** includes a chassis **12**, one or more of front ground engaging mechanisms **13**, and one or more of rear ground engaging mechanisms **14**. The front and rear ground engaging mechanisms **13**, **14** may be wheels or tracks that are in contact with an underlying ground surface and support the chassis **12** above the ground. In the illustrative implementation, the front ground engaging mechanisms **13** are coupled to a front axle **11** that extends laterally, and the rear ground engaging mechanisms **14** are coupled to a rear axle **15** that extends laterally. The front and rear axles **11**, **15** each have respective centerlines, which, in the illustrative implementation, are defined as the axial midpoints of thereof, where the axial direction is shown by the double-headed arrow **114** in FIG. **1**. In the illustrative implementation, the axial direction and the lateral direction are perpendicular to one another. As shown in FIG. **1**, the double headed arrow **116** represents the vertical direction,

which in the illustrative implementation, is perpendicular to the axial and lateral directions.

[0027] In the illustrative implementation, the ground engaging mechanisms **13**, **14** are coupled to the chassis **12** and are configured to rotate to move the agricultural machine **10** in a forward operating direction (which is to the left in FIG. **1**) and in other directions. In some implementations, operation of the agricultural machine **10** is controlled from an operator's cab **16**. The operator's cab **16** may include any number of controls for controlling the operation of the agricultural machine **10**, such as a user interface **220**. In some implementations, operation of the agricultural machine **10** may be conducted by a human operator in the operator's cab **16**, a remote human operator, or an automated system.

[0028] A cutting head **18** is disposed at a forward end of the agricultural machine **10** and is configured to harvest crop and to conduct harvested crop to a slope conveyor **20**. The term harvested crop as used herein includes grain (e.g., corn, wheat, soybeans, rice, oats) and material other than grain (MOG). The cutting head **18** may be a draper, a belt pick-up assembly, a corn head as shown in FIG. **2***a*, a cutting platform with a reel assembly as shown in FIG. **2***b*, or any other cutting head configured to harvest crop in a worksite. Upon receiving the harvested crop from the cutting head **18**, the slope conveyor **20** conducts the harvested crop to a guide drum **22**. The guide drum **22** guides the harvested crop to an inlet **24** of a threshing assembly **26**, as shown in FIG. **1**. In the illustrative implementation, various sub-systems of the agricultural machine **10**, such as the threshing assembly, a clean crop routing assembly **28**, a crop debris routing assembly **60**, and a residue assembly **82**, cooperate to process the harvested crop.

[0029] The threshing assembly **26** includes a housing **34** and one or more threshing rotors. A single threshing rotor **36** is shown in FIG. **1**. The threshing rotor **36** includes a drum **38** arranged along a threshing axis **100**, and the threshing rotor **36** rotates about the threshing axis **100**. The threshing assembly **26** further includes a charging section **40**, a threshing section **42**, and a separating section **44**. The charging section **40** is arranged at a front end of the threshing assembly **26**, the separating section **44** is arranged at a rear end of the threshing assembly **26**, and the threshing section **42** is arranged between the charging section **40** and the separating section **44**. The threshing assembly **26** further includes a thresher basket **43** that is positioned in the threshing section **42** and below the threshing rotor **36**, guide vanes **47** that are positioned above the threshing rotor **36**, and a separating grate **45** that is positioned in the separating section **44** and below the threshing rotor **36**. In the illustrative implementation, the guide vanes **47** guide harvested crop rearwardly through the threshing assembly **26**, and the harvested crop is separated and expands as it engages with the guide vanes **47**. Harvested crop falls through the thresher basket **43** and through the separating grate **45**.

[0030] The harvested crop may be directed to the clean crop routing assembly **28** with a blower **46** and sieves **48**, **50** with louvers. The sieves **48**, **50** can be oscillated axially. The clean crop routing assembly **28** removes MOG and guides grain over a screw conveyor **52** to a grain elevator **94**. The grain elevator **94** deposits the grain in a grain tank **30**, as shown in FIG. **1**. The agricultural machine **10** includes a sensor **230** that is, for example, positioned on the grain elevator **94** and configured to measure a grain yield of the harvested crop. In the illustrative implementation, the yield sensor **230** measures the force of the grain contacting the sensor **230** to determine the yield. The grain in the grain tank **30** can be unloaded by means of an unloading screw conveyor **32** to a grain wagon, trailer, or truck, for example.

[0031] Harvested crop remaining at a rear end of the sieve **50** is again transported to the threshing assembly **26** by a screw conveyor **54** where the harvested crop is reprocessed by the threshing assembly **26**. Harvested crop remaining at a rear end of the sieve **48** is conveyed by an oscillating sheet conveyor **56** to a lower inlet **58** of a crop debris routing assembly **60**. Harvested crop at the threshing assembly **26** is processed by the separating section **44** resulting in straw being separated from other material of the harvested crop. The straw is ejected through an outlet **62** of the threshing assembly **26** and conducted to an ejection drum **64**. The ejection drum **64** interacts with a sheet **66** arranged underneath the ejection drum **64** to move the straw rearwardly. A wall **68** is located to the rear of the ejection drum **64** and guides the straw into an upper inlet **70** of the crop debris routing assembly **60**. In the crop debris routing assembly **60**, blades of a rotatable chopper interact with knives to chop the straw into smaller harvested crop residue.

[0032] Harvested crop residue moves from the crop debris routing assembly **60** to the residue assembly **82** for optional subsequent processing and ejection from the agricultural machine **10**. For example, as shown in FIG. **1**, the residue assembly includes one or more spreaders provided downstream of an outlet **80** of the crop debris routing assembly **60**. One spreader **84** is shown in FIG. **1**. Rotation of blades of the spreader **84** about an axis **88** spreads the chopped straw as the chopped straw exits the agricultural machine **10**. The agricultural machine **10** includes a sensor **212**, such as a camera, that is positioned at the rear end of the agricultural machine **10** and configured capture one or more images of an additional field of view that includes, for example, chopped straw output from the agricultural machine **10** via the residue assembly **82**. It should be appreciated that in some implementations, the additional field of view associated with the sensor **212** is separate from the field of view (e.g., FOV) associated with the at least one sensor **210**, which will be described below.

[0033] It should be appreciated that while an exemplary agricultural machine **10** is described with reference to FIG. **1**, aspects of the disclosure (e.g., the control system and methods herein) are applicable to various agricultural machines configured to harvest crop.

[0034] In some implementations, the agricultural machine **10** includes at least one image sensor **210** configured to capture one or more images of a field of view external to the agricultural machine **10**. In FIG. **1**, the at least one image sensor **210** is embodied as a camera. It should be appreciated that in some implementations, the at least one image sensor **210** may be embodied as one or more cameras (e.g., optical or visual radiation cameras or red, green, blue (RGB) cameras), LiDAR sensors, radar sensors (e.g., long-range terahertz radar, mm wave radar, ultra wideband radar, frequency-modulated continuous wave radar (FMCW), ground penetrating radar), ultrasonic sensors, thermal sensors (e.g., a thermal cameras), stereo cameras, laser vibrometers, infrared nuclear magnetic resonance (NMR) cameras, infrared short-wave infrared (SWIR) cameras, infrared terahertz sensors, or other sensors operable to capture or generate one or more images or data corresponding to the one or more images of the field of view.

[0035] In FIG. **1**, an exemplary field of view is depicted as FOV. In the illustrative implementation, the one or more images of the field of view are one or more images of portions of the worksite, which may include depictions of crop of the type planted in the worksite during the present agricultural cycle (e.g., grain and MOG), crops of other types, soil, debris, and any other material in the worksite. If grain is depicted in the images of the field of view, it may be an indication of harvesting loss (i.e., grain loss during harvesting), pre-harvest loss (i.e., grain loss prior to harvesting), or both. Thus, in some implementations, it is advantageous to locate the field of view at one or more desired locations relative to the agricultural machine **10** to ensure that images of the field of view are indicative of harvesting loss, not pre-harvest loss. Further, in some implementations, it is advantageous to locate the field of view at one or more desired locations relative to particular components of the agricultural machine **10** to ensure that images of the field of view are indicative of harvesting loss associated with the desired components of the agricultural machine **10**, not other components of the agricultural machine **10**. Specifically, in some implementations, it is desirable to determine the harvesting loss that is associated with the cutting head **18**—as opposed to harvesting loss associated with components of the agricultural machine downstream of the cutting head **18**. Further still, in some implementations, it is advantageous to limit the field of view to a specific area to prevent ambient light from reaching the field of view, as ambient light can obscure the images captured by the at least one sensor **210**.

[0036] For at least these reasons, in some implementations, the field of view includes an area rearward of the centerline of the front axle **11**. In some implementations, the field of view is limited to areas rearward of the centerline of the front axle **11**. In some implementations, the field of view includes an area forward of the center line of the rear axle **15**. In some implementations, the field of view is limited to areas forward of the center line of rear axle **15**.

[0037] In some implementations, the field of view includes an area underneath a portion of the chassis **12**, where the term underneath means vertically below and aligned with axially and laterally. In some implementations, as shown in FIG. **1**, the at least one image sensor **210** comprises an image sensor **210***a* located underneath at least one of the threshing assembly **26** and the clean crop routing assembly **28**. In some implementations, the field of view is defined to exclude an area underneath the cutting head **18**. In some implementations, the field of view is defined to exclude area forward of crop cutting devices (e.g., cutting knives) of the cutting head **18**, which, in some implementations, are positioned at a forward end of the cutting head **18**. In some implementations, the field of view is defined to exclude an area underneath or otherwise axially aligned with the slope conveyor **20**, the cutting head **18**, or both. In some implementations, the field of view is defined to exclude an area underneath or otherwise axially aligned with the residue assembly **82**. In some implementations, the field of view is defined to exclude any area configured to receive harvested crop (e.g., straw) output by the residue assembly **82**.

[0038] In some implementations, the at least one image sensor **210***a* is directed vertically downward (i.e., toward the ground) and laterally inward (i.e., toward a lateral midpoint of the agricultural machine **10**) to define the field of view. In the illustrative implementation shown in FIG. **1**, the agricultural machine **10** includes side walls **19**, which are illustrated as transparent to show components of the agricultural machine that are laterally adjacent to the side walls **19**. In the illustrative implementation, the side walls **19** terminate at a lower edge **17**. It should be appreciated that the threshing assembly **26** and the clean crop routing assembly **28** are positioned laterally between opposite side walls **19** of the agricultural machine **10**. As shown in FIG. **1**, in some implementations, the at least one image sensor **210***a* is coupled to one or more of the side walls **19**. While in FIG. **1**, the at least one image sensor **210***a* is positioned at the lower edge **17**, in some implementations, the at least one image sensor **210***a* may be recessed and positioned above the lower edge **17**.

[0039] Referring still to FIG. **1**, in the illustrative implementation, the agricultural machine **10** includes a sensor **214**, such as a camera, positioned on a forward portion of the agricultural machine **10** and configured to capture images associated with an additional field of view, separate from the field of view (e.g., FOV) associated with the at least one sensor **210**. For example, because the sensor **214** is directed to an area forward of the agricultural machine **10**, the sensor **214** is configured to capture images associated with the additional field of view that depicts pre-harvest loss. By contrast, the placement and orientation of the at least one sensor **210***a* of FIG. **1** ensures that the one or more images of the field of view (e.g., FOV) captured by the at least one sensor **210***a* exclude any harvesting loss not associated with the cutting head **18** (e.g., exclusion of pre-harvest loss).

[0040] Referring now to FIGS. **2***a* and **2***b* examples of the cutting head **18** are shown. In FIG. **2***a*, the cutting head **18** is embodied as a corn head **18***a*, and in FIG. **2***b*, the cutting head **18** is embodied as an cutting platform **18***b* with a reel assembly **25**. It should be appreciated that the disclosure is applicable to a draper, a belt pickup, or any other cutting head for harvesting crop in a worksite. In the illustrative implementations, each cutting head **18***a*, **18***b* extends laterally from a first end **21** to a second end **23**, and the at least one image sensor **210** comprises a first sensor **210***b* located on the first end **21** and a second sensor **210***c* located on the second end **23** of the cutting head **18***a*, **18***b*. In the illustrative implementations shown in FIGS. **2***a*, **2***b*, the image sensors **210***b*, **210***c* are directed laterally inward toward the field of view. In the illustrative implementation, the field of view includes an area forward of the centerline of the front wheel axle **11**, such as an area axially aligned with the cutting head **18**, an area axially aligned with the slope conveyor **20**, or both. In contrast, in implementations associated with the at least one image sensor **210***a* of FIG. **1**, the field of view (e.g., FOV) may exclude any area axially aligned with the cutting head **18**, axially aligned the slope conveyor **20**, or both, for example, to ensure that images from the field of view captured by the at least one image sensor **210***c* exclude harvesting loss that is not associated with the cutting head **18** (e.g., exclusion of pre-harvest loss).

[0041] In some implementations in which the at least one image sensor **210** comprises a first sensor **210***b* located on the first end **21** and a second sensor **210***c* located on the second end **23** of the cutting head **18***a*, **18***b*, the image sensors **210***b*, **210***c* are directed laterally inward toward a field of view that includes an area rearward of the centerline of the front axle **11**. In some implementations in which the at least one image sensor **210** comprises a first sensor **210***b* located on the first end **21** and a second sensor **210***c* located on the second end **23** of the cutting head **18***a*, **18***b*, the image sensors **210***b*, **210***c* are directed laterally inward toward a field of view that is limited to an area rearward of the centerline of the front axle **11**.

[0042] Referring again to FIG. **1**, in the illustrative implementation, the agricultural machine **10** includes at least one light emitting device **216** configured to emit light into the field of view. In the illustrative implementation, the at least one light emitting device **216** comprises at least one of: one or more lasers (e.g. infrared lasers, ultraviolet lasers, x-ray lasers, gamma-ray lasers, or any other laser configured to emit light into the field of view); one or more light emitting diodes (LEDs) (e.g., single-color LEDs, white LEDs, RGB LEDs, phosphor-based LEDs, mixed white LEDs, other white LEDs, Perovskite light-emitting diodes (PeLEDs) or any other LED array of any type configured to emit light into the field of view; one or more incandescent bulbs, halogen bulbs, or compact fluorescent lamps; or any other type of light emitter configured to emit light into the field of view. The at least one light emitting device **216** may be coupled to the chassis **12**, the side walls **19**, or positioned elsewhere on the agricultural machine **10** and directed toward the field of view. In some implementations, the at least one light emitting device **216** includes multiple light sources spaced apart from each other such that the at least one light emitting device **216** is configured to emit light from a plurality of locations into the field of view. In some implementations, the at least one light emitting device **216** is configured to emit light in a plurality of ranges of wavelengths. Each range includes one or more wavelengths of light. In some implementations, one or more ranges of wavelengths of the plurality of ranges of wavelengths are associated with one or more specific colors of visible light (e.g., red, orange, yellow, green, blue, violet). In some implementations, the at least one light emitting device **216** is configured to emit white light. In some implementations, the at least one light emitting device **216** is configured to emit light in a plurality of intensities.

[0043] In some implementations, the agricultural machine **10** includes at least one light filter configured to filter the emitted light captured by the at least one image sensor **210**. In some implementations, the at least one light filter of the agricultural machine **10** is at least one polarizer **218** configured to filter light being received by the at least one image sensor **210** into one or more beams of light with adjusted (e.g., increased) polarization. In the illustrative implementation, the at least one polarizer **218** is embodied as at least one of: one or more linear polarizers (e.g., absorptive polarizers, beam-splitter polarizers) and one or more circular polarizers. In an illustrative implementation, the at least one polarizer may be embodied as a fiber polarization controller (e.g., coiled fiber or squeezed fiber). In some implementations, as shown in FIG. **1**, the at least one light filter (e.g., the at least one polarizer **218**) may be coupled to or otherwise positioned adjacent to the at least one light emitting device **216** (e.g. in the path of the emitted light) to polarize the wavelengths of light emitted by the at least one light emitting device **216**. In some implementations, as shown in FIG. **1**, the at least one light filter (e.g., the at least one polarizer **218**) may be coupled to or otherwise positioned adjacent to the at least one image sensor **210** (e.g. in the path of the received light) to filter the light being received by the at least one image sensor **210**.

[0044] Referring now to FIG. **3**, an example control system **200** is shown. The control system **200** includes one or more memories **208** included in or accessible by the controller **202** and one or more processors **206** included in or accessible by the controller **202**. The one or more processors **206** are configured to execute instructions (e.g., one or more algorithms) stored on the one or more memories **208**. The controller **202** may be a single controller or a plurality of controllers operatively coupled to one another. The controller **202** may be positioned on the agricultural machine **10** or positioned remotely, away from the agricultural machine **10**. The controller **202** may be coupled via a wired connection or wirelessly to other components of the agricultural machine **10** and to one or more remote devices. In some instances, the controller **202** may be connected wirelessly via Wi-Fi, Bluetooth, Near Field Communication, or another wireless communication protocol to other components of the agricultural machine **10** and to one or more remote devices.

[0045] Referring still to FIG. **3**, in the illustrative implementation, the controller **202** is operatively coupled to at least one characteristic sensor **204**, the at least one image sensor **210**, the sensor **212**, the sensor **214**, and the sensor **230**. In the illustrative implementation, the controller **202** is configured to receive data corresponding to the one or more images of the field of view (e.g. FOV) from the at least one image sensor **210** and data corresponding to the one or more images of the additional fields of view from the image sensors **212**, **214**. In some implementations, the controller **202** includes or is operatively coupled to a spectrophotometer configured to measure reflectance of material in the field of view (e.g., FOV).

[0046] In the illustrative implementation, the controller **202** is operatively coupled to the at least one light emitting device **216** and configured to send one or more signals to the at least one light emitting device **216** causing the at least one light emitting device **216** to emit light into the field of view at a plurality of different ranges of wavelengths, a plurality of different intensities, from different types of light emitters (e.g. Laser, LEDs), or from a different light source location. In some implementations, the controller **202** is operatively coupled to the at least one light filter and configured to send one or more signals to the at least one light filter causing the at least one light filter to filter the light received by the at least one image sensor **210**. In some implementations, the light filter is adjusted between ON/OFF modes, and in some implementations, the degree or amount of light being filtered is adjusted. In some implementations, the controller **202** is operatively coupled to the at least one polarizer **218** and configured to send one or more signals to the at least one polarizer **218** causing the at least one polarizer **218** to modify the polarization state of the light received by the at least one image sensor **210**.

[0047] Referring still to FIG. **3**, in some implementations, the controller **202** is operatively coupled to a display, for example, the display **222** of the user interface **220**, and configured to send one or more signals to the display based on a determination of an amount of grain shown in one or more images of the field of view. In the illustrative implementation, the controller **202** is operatively coupled to the user interface **220** and configured to receive one or more signals from the user interface **220**, which may include inputs for operational characteristics of the agricultural machine **10** (e.g., speed of the agricultural machine **10**, direction of the travel of the agricultural machine **10**, operational characteristics of the cutting head **18**, operational characteristics of the threshing assembly **26**, operational characteristics of the clean crop routing assembly **28**, operational characteristics of the crop debris routing assembly **60**, and operational characteristics of the residue assembly **82**. Operational characteristics of the cutting head **18** include cutting head angle along axial span thereof, cutting head height relative to ground, cutting head speed, reel speed, reel position, reel tine angle, corn head deck plate spacing, draper belt speed, cutting head down force, and lateral tilt of cutting head. Operational characteristics of the threshing assembly **26** include threshing rotor speed, position of the thresher basket relative to the threshing rotor, and guide vane orientation. Operational characteristics of the clean crop routing assembly **28** include blower speed and sieve position. Operational characteristics of the crop debris routing assembly **60** include chopper speed and position of the knives relative to the chopper. Operational characteristics of the residue assembly **82** include spreader speed and spreader orientation.

[0048] In some implementations, the operational characteristics of the agricultural machine **10** are input by a user via the user interface **220** based on the displayed determination of the amount of grain shown in the one or more images of the field of view. It should be appreciated that in some implementations, the controller **202** is configured to adjust one or more of the operational characteristics of the agricultural machine **10** automatically (e.g., based on a determination of an amount of grain shown in one or more images of the field of view and without instruction from the user interface **220**).

[0049] In some implementations, the controller **202** is configured to receive one or more signals indicative of one or more environmental characteristics of a worksite in which the harvesting operation is being performed by the agricultural machine **10**. In the illustrative implementation, the environmental characteristics comprise one or more wind characteristics (e.g., speed, direction), one or more sunlight characteristics (e.g., amount, intensity, direction, wavelength), one or more plant residue characteristics (e.g., plant residue color, plant residue volume, plant residue arrangement), and one or more soil characteristics (e.g., moisture level, nutrient levels, pH, texture, compaction). As used herein plant residue includes MOG and plant material not intended for harvesting. In some implementations, the controller **202** is configured to receive one or more signals indicative of one or more crop characteristics of a worksite in which the harvesting operation is being performed by the agricultural machine **10**. In the illustrative implementation, the crop characteristics comprise crop type (e.g., corn, wheat, soybeans), constituent levels (e.g., moisture levels, protein levels, starch levels, fiber levels, sugar levels, oil levels), standing crop color, standing crop size, down crop values, grain shape, grain size, grain color. In some implementations, the one more environmental characteristics, the one or more crop characteristics, or both may be received via the user interface **220**, accessed via private or public data stores associated with the worksite (e.g., Internet-based), accessed from the memory **208**, received via at least one characteristic sensor **204**, or received via another sensor associated with the agricultural machine **10**. In illustrative implementations, the at least one characteristic sensor **204** may be located on or away from the agricultural machine **10**, and is configured to measure one or more environmental characteristics associated with the worksite, one or more crop characteristics associated with the worksite, or both.

[0050] In some implementations, the agricultural machine **10** is included in a fleet of agricultural machines, which may operate in the same worksite as the agricultural machine **10** or different worksite and at the same time as the agricultural operation performed by the agricultural machine **10** or at another time. In some implementations, the controller **202** is configured to receive an indication of at least one environmental characteristic, at least one crop characteristic, or both from another agricultural machine, which, for example, may be included in the fleet.

[0051] In some implementations, for example, to initiate an agricultural operation in a worksite, the controller **202** is configured to cause the at least one light emitting device **216** to emit light into the field of view at a selected range of wavelengths based on at least one environmental characteristic, at least one crop characteristic, or both. In some implementation, the controller **202** is configured to select an expressed value of another light emission variable (e.g., the intensity of light emitted by the at least one light emitting device **216**, the filter degree or amount (e.g., polarization state of the light), the type of light emitter of the light emitting device **216** that emits light, and the location of one or more active light sources of the at least one light emitting device **216**. In some implementations, the controller **202** is configured to cause the at least one light emitting device **216** to emit light into the field of view at a selected range of wavelengths based on the received indication of at least one environmental characteristic, at least one crop characteristic, or both from the another agricultural machine. In some implementations, the controller **202** is configured to select an expressed value of another light emission variable based on the received indication of at least one environmental characteristic, at least one crop characteristic, or both from the another agricultural machine.

[0052] Referring still to FIG. **3**, in some implementations, the controller **202** is operatively coupled to at least one cutting head actuator **224**, at least one ground engaging mechanism actuator **226**, or both. The at least one cutting head actuator **224** and the at least one ground engaging mechanism actuator **226** may each be embodied as at least one of a control valve, a motor, a linear (e.g., cylindrical) actuator, a rotary actuator, or another actuator configured to cause adjustment of an operational characteristic of the agricultural machine **10**. As shown in FIG. **3**, in some implementations, the controller **202** is operatively coupled to an engine **228** of the agricultural machine and configured send one or more signals to the engine **228** to adjust the speed of the agricultural machine **10**. In some implementations, the controller **202** is operatively coupled to one or more additional sub-system actuators **225** (e.g., control valves, motors, linear actuators, rotary actuators), configured to cause adjustment of one or more operational characteristics the threshing assembly **26**, the clean crop routing assembly **28**, the crop debris routing assembly **60**, and the residue assembly **82**.

[0053] In an example method **300** that is shown in FIG. **4**, the control system **200** is usable to determine a grain loss indicator for a portion of a worksite being harvested. In some implementations, the grain loss indicator comprises at least one of a grain loss amount, a grain loss as a percentage of yield, and a monetary value of grain loss.

[0054] Referring to FIG. **4**, in some implementations, at a block **302**, the at least one light emitting device **216** emits light into the field of view (e.g. FOV). In some implementations, at a block **303**, at least one light filter filters the light after emission. In some implementations, at least one polarizer **218** filters light into one or more beams of light with adjusted (e.g., increased) polarization. At a block **304**, the at least one image sensor **210** captures one or more images of the field of view. In some implementations, the one or more captured images are polarized. At a block **306**, the controller **202** receives data corresponding to one or more images of the field of view from the at least one image sensor **210**. At a block **308**, the controller **202** determines one or more reflectance values based on the received data corresponding to the one or more images. In the illustrative implementation, each reflectance value determined by the controller **202** is associated with a portion of an image of the one or more images of the field of view. At a block **310**, the controller **202** determines an amount of grain shown in the one or more images of the field of view based on the one or more determined reflectance values. In some implementations, at a block **312**, the controller **202** determines the grain loss indicator (e.g., grain loss amount, grain loss as a percentage of yield, and monetary value of grain loss) for a portion of a worksite being harvested based on the amount of grain shown in the one or more images of the field of view, the area of the portion of the worksite being harvested, and the area of the field of view. In some implementations, at a block **314**, the controller **202** causes adjustment of one or more operational characteristics of the agricultural machine **10** based on the grain loss indicator. For example, the controller **202** sends one or more signals to at least one of the at least one cutting head actuator **224**, the at least one ground engaging mechanism actuator **226**, and the engine **228** to cause adjustment of an associated operational characteristic. For example, in some implementations, the controller **202** compares the grain loss indicator to a threshold value for grain loss (which may be established, e.g., in part, based on one or more environmental characteristics associated with the worksite, one or more crop characteristics associated with the worksite, or both), and the controller **202** sends the one or more signals to at least one of the at least one cutting head actuator **224**, the at least one ground engaging mechanism actuator **226**, and the engine **228** if the grain loss indicator exceeds the threshold value for grain loss.

[0055] In some implementations, the controller **202** is configured to determine the area of the portion of the worksite being harvested (e.g., based on lateral length of cutting head from first end **21** to second end **23** and at least one of: (i) distance traveled by agricultural machine **10** while harvesting the portion of the worksite and (ii) speed of agricultural machine **10** and time elapsed while harvesting the portion of the worksite). In some implementations, the controller **202** is configured to determine the area of the field of view (e.g., based on information accessed from memory **202**, received via user interface **220**, or received via the at least one image sensor **210**). In some implementations, the controller **202** is configured to determine the grain loss amount based on the amount of grain shown in the one or more images of the field of view, the area of the portion of the worksite being harvested, and the area of the field of view.

[0056] In some implementations, the controller **202** is configured to obtain a yield value (e.g., bushels per acre) for the portion of the worksite being harvested (e.g., based on one or more signals received from the yield sensor **230**). In some implementations, the controller **202** is configured to determine the grain loss as a percentage of yield based on the grain loss amount for the portion of the worksite being harvested and the yield value for the portion of the worksite being harvested.

[0057] In some implementations, the controller **202** is configured to obtain a market value of grain (e.g., dollars per bushels) of the crop type being harvested. In some implementations, the controller **202** is configured to determine the monetary value of grain loss based on the grain loss amount for the portion of the worksite being harvested and the market value of grain.

[0058] In some implementations, at a block **316**, the controller **202** receives one or more signals indicative of one or more crop characteristics associated with the worksite. In some implementations, the controller **202** determines the amount of grain shown in the one or more images of the field of view further based on one or more received crop characteristics associated with the worksite. In some implementations, at a block **318**, the controller **202** receives one or more signals indicative of one or more environmental characteristics associated with the worksite. In some implementations, the controller **202** determines the amount of grain shown in the one or more images of the field of view further based on one or more received environmental characteristics associated with the worksite. For example, based on the environmental characteristics and crop characteristics, the appearance of grain in the one or more images of the field of view (and the corresponding data associated therewith) may differ such that evaluating the environmental characteristics and crop characteristics is advantageous for accurately determining the amount of grain shown in the one or more images of the field of view. For example, a change in sunlight amount may cause grain to appear a slightly different color.

[0059] In some implementations, the controller **202** determines the amount of grain shown in the one or more images of the field of view based on one or more relationships between reflectance values and grain. For example, in some implementations, the controller **202** compares the determined reflectance value to predetermined reflectance values of one or more grain types (e.g., stored in the memory **208**) to determine whether the determined reflectance value indicates the presence of grain in the one or more images of the field of view. Crop characteristics and environmental characteristics affect the relationship between the reflectance values and the grain. For example, the predetermined reflectance values of the one or more grain types may differ based on the presence, absence, or degree of one or more crop characteristics, one or more environmental characteristics, or both. For example, a change in sunlight amount may cause grain to have a different reflectance value.

[0060] In some implementations, the control system **200** leverages machine learning to more accurately determine the amount of grain shown in the one or more images of the field of view. For example, the controller **202** may rely on data from other harvesting operations, other agricultural machines, or other worksites (e.g., where the same crop type is harvested) to arrive at predetermined reflectance values of grain. Thus, as shown in FIG. **4**, at a block **320**, the controller **202** is configured to compare the data corresponding to the one or more captured images to data corresponding to one or more additional images including grain of a crop type associated with the worksite, and at a step **322**, the controller **202** is configured to determine the amount of grain shown in the one or more images of the field of view further based on comparison between the data corresponding to the one or more captured images and the data corresponding to the one or more additional images.

[0061] Referring now to FIG. **5**, in an example method **400**, the control system **200** is usable to perform a calibration operation as to the range of wavelengths of light. And subsequently, the controller **202** determines the grain loss indicator for the portion of the worksite being harvested at the calibrated range of wavelengths of light. For example, at a block **402**, the controller **202** sends one or more signals to the at least one light emitting device **216** causing the at least one light emitting device **216** to emit light into the field of view in a plurality of ranges of wavelengths. In some implementations, at a block **403**, at least one light filter filters the light after emission. In some implementations, light from the field of view may be polarized by the at least one polarizer **218**. At a block **404**, the at least one image sensor **210** captures one or more images of the field of view for each range of wavelengths. In some implementations, the one or more captured images are polarized. At a block **406**, the controller **202** receives data corresponding to one or more images of the field of view from the at least one image sensor **210** for each range of wavelengths. At a block **408**, the controller **202** determines, for each range, one or more reflectance values based on the received data corresponding to the one or more images. In the illustrative implementation, each reflectance value is associated with a portion of an image of the one or more images. At a block **410**, the controller **202** determines, for each range, an amount of grain shown in the one or more images of the field of view based on the one or more determined reflectance values. At a block **412**, the controller **202** causes the at least one light emitting device **216** to emit light into the field of view in the range of wavelengths associated with the greatest amount of grain determined, by the controller **202**, to be shown in the one or more images.

[0062] It should be appreciated that, in the method **400**, in some implementations, the controller **202** receives one or more crop characteristics and one or more environmental characteristics and determines, for each range of wavelengths, the amount of grain shown in the one or more images of the field of view further based on the one or more received crop characteristic and one or more environmental characteristics. It should also be appreciated that, in the method **400**, in some implementations, the controller **202** leverages machine learning, as described in blocks **320**, **322**, to more accurately determine, for each range of wavelengths, the amount of grain shown in the one or more images of the field of view. Subsequent to execution of the calibration method **400**, the control system **200** executes blocks associated with the method **300** using the range of wavelengths determined via the calibration method **400**.

[0063] The range of wavelengths emitted by the at least one light emitting device **216** is an example of a light emission variable. Other light emission variables includes: the intensity of light emitted by the at least one light emitting device **216**, the filter amount or degree (e.g., polarization state of the light, which, in some implementations is adjustable by the at least one polarizer **218**), the type of light emitter of the light emitting device **216** (e.g., LEDs, Laser, or both) that emits light, and the location of one or more active light sources of the at least one light emitting device **216**. The method **400** is one example of calibrating a light emission variable. In some implementations, in addition to calibrating the range of wavelengths of light, the control system **200** (in an example method **500**) is useable to calibrate other light emission variables.

[0064] For example, as shown in FIG. **6**, for each light emission variable, at a block **502**, the controller **202** sends one or more signals to the at least one light emitting device **216** (or the at least one light filter, e.g., the at least one polarizer **218** in the case of the polarization state of the light) to cause the light emission variable to be expressed at a plurality of different values (e.g., different intensities, filter amount or degree (e.g., different polarization states), different types of light emitters, different location of light source). In some implementations, at a block **503**, at least one light filter filters the light after emission. In some implementations, light from the field of view may be polarized by the at least one polarizer **218**. At a block **504**, the at least one image sensor **210** captures one or more images of the field of view for each expressed value of the light emission variable. In some implementations, the one or more captured images are polarized. At a block **506**, the controller **202** receives data corresponding to one or more images of the field of view from the at least one image sensor **210** for each expressed value of the light emission variable. At a block

508, the controller **202** determines, for each expressed value of the light emission variable, one or more reflectance values based on the received data corresponding to the one or more images. At a block **510**, the controller **202** determines, for each expressed value of the light emission variable, an amount of grain shown in the one or more images of the field of view based on the one or more determined reflectance values. At a block **512**, the controller **202** causes the at least one light emitting device **216** (or the at least one light filter, e.g., the at least one polarizer **218** in the case of the polarization state of the light) to express the light emission variable at the expressed value associated with the greatest amount of grain determined, by the controller **202**, to be shown in the one or more images.

[0065] It should be appreciated that, in the method **500**, in some implementations, the controller **202** receives one or more crop characteristics and one or more environmental characteristics and determines, for each expressed value of the light emission variable, the amount of grain shown in the one or more images of the field of view further based on the one or more received crop characteristic and one or more environmental characteristics. It should also be appreciated that, in the method **500**, in some implementations, the controller **202** leverages machine learning, as described in blocks **320**, **322**, to more accurately determine, for each expressed value of the light emission variable, the amount of grain shown in the one or more images of the field of view. Subsequent to execution of the calibration method **500**, the control system **200** executes blocks associated with the method **300** using the expressed value of the light emission variable determined via the calibration method **500**.

[0066] In some implementations, the controller **202** is configured to receive one or more signals from the user interface **220** indicative of a selected value for one or more of the light emission variables. In such implementations, the controller **202** is configured to cause the at least one light emitting device **216** or the at least one light filter (e.g., the at least one polarizer **218**) to express the corresponding one or more light emission variables at the selected value. It should be appreciated that, in some implementations, the control system **200** executes blocks associated with the method **300** using the selected value for the one or more light emission variables.

[0067] In some implementations, the controller **202** is configured to determine a desired range of wavelengths from the plurality of ranges of wavelengths based on at least one environmental characteristic, at least one crop characteristic, or both. In some implementations, the desired range of wavelengths is the range of wavelengths that, when emitted, results in the greatest amount of grain being visible (e.g., by an operator) or determined (e.g., by the controller **202**) to be shown in the one or more images of the field of view; in some implementations, the desired range of wavelengths is the range of wavelengths that, when emitted, results in the most accurate amount of grain being determined (e.g., by the controller **202**) to be shown in the one or more images of the field of view; in some implementations, the desired range of wavelengths is based on both the greatest amount of grain determined (e.g., by the controller **202**) to be shown and the most accurate amount of grain determined to be shown in the one or more images of the field of view. For example, the controller **202** may determine the desired range of wavelengths based: (1) receipt of at least one environmental characteristic, at least one crop characteristic, or both, and (2) data (e.g., stored on in memory **208**) representing a relationship between: (i) the at least one environmental characteristic, the at least one crop characteristic, or both, (ii) the plurality of ranges of wavelengths, and (iii) the amount of grain determined to be shown in the one or more images of the field of view. In such implementations, the controller **202** is configured to cause the at least one light emitting device **216** to emit light into the field of view at the desired range of wavelengths.

[0068] It should be appreciated that, in some implementations, the controller **202** is configured to determine a desired value for each light emission variable based on at least one environmental characteristic, at least one crop characteristic, or both. In some implementations, the desired value for each light emission is the value that, when expressed, results in the greatest amount of grain being visible (e.g., by an operator) or determined (e.g., by the controller **202**) to be shown in the one or more images of the field of view; in some implementations, the desired value for each light emission variable is the value that, when expressed, results in the most accurate amount of grain being determined (e.g., by the controller **202**) to be shown in the one or more images of the field of view; in some implementations, the desired value for each light emission variable is based on both the greatest amount of grain determined (e.g., by the controller **202**) to be shown and the most accurate amount of grain determined to be shown in the one or more images of the field of view. For example, the controller **202** may determine the desired value for each light emission variable based: (1) receipt of at least one environmental characteristic, at least one crop characteristic, or both, and (2) data (e.g., stored on in memory **208**) representing a relationship between: (i) the least one environmental characteristic, the at least one crop characteristic, or both, (ii) the expressed values of the light emission variables, and (iii) the amount of grain determined to be shown in the one or more images of the field of view. In such implementations, the controller **202** is configured to cause the at least one light emitting device **216** to emit light into the field of view at the desired range of wavelengths.

[0069] Referring now to FIG. **7**, in an example method **600**, the control system **200** is usable to determine grain loss occurring during the harvesting operation, which is exclusive of pre-harvest grain loss. At a block **602**, the controller **202** receives an indication of pre-harvest grain loss (e.g., via the sensor **214**, via the user interface **220**, via private or public data stores (e.g., Internet-based) associated with the worksite). At a block **604**, the controller **202** determines the pre-harvest grain loss for the portion of the worksite being harvested. As described with reference to example method **300**, at block **312**, (based on the grain amount determined at the block **310**), the controller **202** determines the grain loss indicator for the portion of the worksite being harvested. In the illustrative implementation, at a block **606**, the controller **202** determines the cutting head grain loss indicator based on the determined pre-harvest grain loss for the portion of the worksite being harvested and the determined grain loss indicator for the portion of the worksite being harvested. Similar to the grain loss indicator, in some implementations, the cutting head grain loss indicator is expressed as a grain loss amount during harvesting, grain loss during harvesting as a percentage of yield, and a monetary value of grain loss during harvesting. It should be appreciated that the cutting head grain loss indicator is associated with the grain loss resulting from operation of the cutting head **18**.

[0070] In some implementations, at a block **608**, the controller **202** sends one or more signals to a display operatively coupled to the controller **202** (e.g., display **222**) causing the cutting head grain loss indicator to be displayed. In some implementations, at a block **610**, the controller **202** generates a cutting head grain loss map, which indicates one or more cutting head grain loss indicators at one or more corresponding locations in the worksite. For example, the one or more cutting head grain loss indicators may be color-coded based on degree of grain loss at corresponding locations of the worksite. In some implementations, the controller **202** causes the cutting head grain loss map to be a displayed on a display operatively coupled to the controller **202** (e.g., display **222**). In some implementations, at a block **612**, the controller **202** causes adjustment of one or more operational characteristics of the agricultural machine **10** based on the cutting head grain loss indicator. For example, the controller **202** sends one or more signals to at least one of the at least one cutting head actuator **224**, the at least one ground engaging mechanism actuator **226**, and the engine **228** to cause adjustment of associated operational characteristic. For example, in some implementations, the controller **202** compares the cutting head grain loss indicator to a threshold value for cutting head grain loss (which may be established, e.g., based on one or more environmental characteristics associated with the worksite, one or more crop characteristics associated with the worksite, or both), and the controller **202** sends the one or more signals to at least one of the at least one cutting head actuator **224**, the at least one ground engaging mechanism actuator **226**, and the engine **228** if the cutting head grain loss indicator exceeds the threshold value for cutting head grain loss.

[0071] While the disclosure has been illustrated and described in detail in the drawings and foregoing description, such illustration and description is to be considered as exemplary and not restrictive in character, it being understood that illustrative implementation(s) have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected. It will be noted that alternative implementations of the present disclosure may not include all of the features described yet still benefit from at least some of the advantages of such features. Those of ordinary skill in the art may readily devise their own implementations that incorporate one or more of the features of the present disclosure and fall within the spirit and scope of the present disclosure as defined by the appended claims.

Claims

1. An agricultural machine for reducing harvesting loss during a harvesting operation, comprising: front ground engaging mechanisms coupled to a front axle; rear ground engaging mechanisms coupled to a rear axle; a chassis supported above a surface by the front ground engaging mechanisms and rear ground engaging mechanisms; a cutting head located forward of the front ground engaging mechanisms and configured to harvest crop in a worksite; at least one image sensor configured to capture one or more images of a field of view, the field of view being external to the agricultural machine and including an area rearward of a centerline of the front axle; and a controller configured to receive data corresponding to one or more images of the field of view from the at least one image sensor, determine the amount of grain shown in the one or more images of the field of view based on the received data, and adjust at least one operational characteristic of the agricultural machine based on the amount of grain determined to be shown in the one or more images of the field of view.
2. The agricultural machine of claim 1, wherein the field of view includes an area underneath a portion of the chassis.
3. The agricultural machine of claim 1, further comprising: at least one light emitting device configured to emit light into the field of view.
4. The agricultural machine of claim 3, wherein the at least one light emitting device includes at least one laser.
5. The agricultural machine of claim 1, wherein the controller is configured to adjust at least one of the following operational characteristics of the agricultural machine based on the amount of grain determined to be shown in the one or more images of the field of view: speed of the agricultural machine, direction of travel of the agricultural machine, at least one operational characteristic of the cutting head, and at least one operational characteristic of a threshing assembly of the agricultural machine that is configured to process crop harvested by the cutting head.
6. The agricultural machine of claim 1, wherein the field of view includes an area forward of a center line of the rear axle.
7. The agricultural machine of claim 1, further comprising: a threshing assembly positioned rearward of the cutting head; and a clean crop routing assembly configured to cooperate with the threshing assembly to separate grain from material other than grain of the harvested crop; wherein the at least one image sensor is located underneath at least one of the threshing assembly and the clean crop routing assembly.
8. The agricultural machine of claim 1, wherein the at least one image sensor is directed laterally inward toward a lateral centerline of the agricultural machine.
9. The agricultural machine of claim 8, further comprising: a threshing assembly positioned rearward of the cutting head; a clean crop routing assembly configured to cooperate with the threshing assembly to separate grain from material other than grain of the harvested crop; and side walls between which the threshing assembly and the clean crop routing assembly are positioned; wherein the at least one image sensor is located on at least one of the side walls.
10. The agricultural machine of claim 8, wherein the cutting head extends laterally from a first end

to a second end; and wherein the at least one image sensor is located on at least one of the first end and the second end of the cutting head.

11. The agricultural machine of claim 1, further comprising: a threshing assembly positioned rearward of the cutting head; a clean crop routing assembly configured to cooperate with the threshing assembly to separate grain from material other than grain of the harvested crop; a spreader configured to output the material other than grain from the agricultural machine; and an additional image sensor configured to capture one or more images of an additional field of view that includes an area configured to receive the material other than grain of the harvested crop output from the agricultural machine.

12. The agricultural machine of claim 1, wherein the controller is configured to: receive an indication of a pre-harvest grain loss; and generate a cutting head harvesting loss map, which indicates grain loss during the harvesting operation relative to locations in the worksite, based on the indication of pre-harvest grain loss and based on the amount of grain determined to be shown in the one or more images of the field of view.

13. An agricultural machine for reducing harvesting loss during a harvesting operation, comprising: front ground engaging mechanisms configured to rotate during movement of the agricultural machine; rear ground engaging mechanisms configured to rotate during movement of the agricultural machine; a chassis supported above a surface by the pair of front ground engaging mechanisms and the pair of rear ground engaging mechanisms; a cutting head located forward of the pair of front ground engaging mechanisms and configured to harvest crop; a slope conveyor located rearward of the cutting head and forward of an inlet of a threshing assembly, the threshing assembly being configured to process harvested crop; at least one image sensor configured to capture one or more images of a field of view that is external to the agricultural machine and includes an area underneath the threshing assembly; and a controller configured to receive data corresponding to one or more images of the field of view from the at least one image sensor and determine the amount of grain shown in the one or more images of the field of view based on the received data corresponding to one or more images of the field of view.

14. The agricultural machine of claim 13, wherein the controller is configured to adjust at least one operational characteristic of the cutting head based on the amount of grain shown in the one or more images of the field of view.

15. The agricultural machine of claim 13, wherein the controller is configured to adjust at least one operational characteristic of the threshing assembly based on the amount of grain shown in the one or more images of the field of view.

16. The agricultural machine of claim 13, wherein the field of view includes an area forward of the rear ground engaging mechanisms.

17. The agricultural machine of claim 13, further comprising: a clean crop routing assembly configured to cooperate with the threshing assembly to separate grain from material other than grain of the harvested crop; and a first side wall and a second side wall between which the threshing assembly and the clean crop routing assembly are positioned; wherein the at least one image sensor is located on at least one of the first side wall and the second side wall.

18. The agricultural machine of claim 13, wherein the controller is configured to: receive an indication of a pre-harvest grain loss; and determine a cutting head grain loss value based on the indication of pre-harvest grain loss and based on the amount of grain determined to be shown in the one or more images of the field of view.

19. A method for reducing harvesting loss of an agricultural machine during a harvesting operation, the method comprising: capturing one or more images of a field of view that is external to the agricultural machine and includes an area underneath a threshing assembly of the agricultural machine, the threshing assembly being configured to process harvested crop received from a slope conveyor located rearward of a cutting head that harvests crop and forward of an inlet of the threshing assembly; receiving, via a controller, data corresponding to one or more images of the field of view from the at least one image sensor; and determining, via the controller, the amount of grain shown in the one or more images of the field of view based on the received data corresponding to one or more images of the field of view.

20. The method of claim 19, further comprising: adjusting at least one operational characteristic of the agricultural machine based on the amount of grain determined, via the controller, to be shown in the one or more images of the field of view.